

2.3: Risk and Volatility in Financial Time Series



In the previous section, you learnt that financial time series have distinct statistical properties. You saw that prices are often non-stationary, while returns tend to be stationary.

This section introduces volatility as the central quantitative measure of financial risk and explains how time series behaviour shapes how risk is measured, interpreted, and managed.

Recall, in finance, risk refers to uncertainty about future outcomes, particularly uncertainty about returns.

Two investments may have the same average return but very different risk profiles. One may deliver relatively stable outcomes, while the other may experience large swings. Risk captures this variability.

What Is Volatility?

Volatility is the quantitative expression of financial risk. It captures how much returns fluctuate around their average value over time and therefore measures the degree of uncertainty faced by an investor.

In practical terms, volatility answers the question:

How unpredictable are investment outcomes?

An asset with low volatility tends to deliver returns that are relatively stable and predictable. An asset with high volatility experiences larger and more frequent swings, making outcomes harder to anticipate. Importantly, volatility does not describe whether returns are positive or negative. It describes how dispersed they are.

This distinction matters because two investments can have the same average return but very different risk profiles. One may achieve its return smoothly, while the other does so through sharp

gains and losses. Volatility captures this difference and is therefore central to portfolio construction, risk management, and financial decision-making.

In modern finance, volatility is not treated as a fixed characteristic. Instead, it is understood to evolve over time, responding to new information, market stress, and changes in economic conditions.

Volatility as a Dynamic Quantity

Financial markets do not experience constant risk. Periods of relative calm are interrupted by episodes of heightened uncertainty, such as during financial crises, geopolitical shocks, or major policy announcements.

This means that volatility itself varies over time. Risk today may be very different from risk next month, even if expected returns remain similar. Recognising volatility as a dynamic quantity allows analysts and decision-makers to adjust exposure, capital allocation, and hedging strategies in response to changing market conditions.

Volatility Clustering

A key empirical feature of financial markets is **volatility clustering**.

Volatility clustering refers to the tendency for periods of high volatility to be followed by further high volatility, and periods of low volatility to be followed by continued calm. In other words, volatility is persistent.

This does not mean that returns themselves are predictable in direction. Price changes may still be positive or negative in an unpredictable way. What *is* predictable is the *magnitude* of fluctuations.

In intuitive terms:

- turbulent markets tend to remain turbulent for a while
- calm markets tend to remain calm for a while

This behaviour has been observed consistently across asset classes, markets, and time periods.

Examples of Volatility Clustering

Volatility clustering is not an abstract statistical idea. It is visible in major financial events:

- **The Global Financial Crisis (2008–2009)**

After the collapse of Lehman Brothers, markets did not simply experience one large shock and return to normal. Instead, extreme volatility persisted for months as uncertainty about financial institutions and economic stability remained high.

- **The COVID-19 Market Shock (2020)**

Equity markets experienced a sudden surge in volatility during the initial pandemic response. Even after prices began to recover, volatility remained elevated as markets adjusted to ongoing uncertainty about lockdowns, stimulus, and global growth.

These episodes illustrate that once markets enter a high-risk state, uncertainty tends to feed on itself.

Why Volatility Clustering Matters for Risk Management

Volatility clustering has important implications for how risk is measured and managed.

If volatility were constant, a single historical estimate would be sufficient. But because volatility changes over time, risk measures must adapt to current market conditions.

This affects:

- **portfolio risk management**, where position sizes and diversification depend on current risk levels
- **derivatives pricing**, where expected future volatility directly influences option values
- **regulatory capital requirements**, where banks must hold more capital during periods of elevated risk
- **corporate hedging decisions**, where the cost and necessity of hedging vary with market uncertainty

Ignoring volatility clustering leads to systematic underestimation of risk during stressed periods and overconfidence during calm periods.

Linking Back to Time Series Behaviour

The concepts introduced in the previous section now come together:

- Returns are approximately stationary, making their variance meaningful
 - Volatility measures the variability of returns
 - Volatility itself exhibits persistence over time
 - This persistence allows risk to be modelled, monitored, and managed dynamically
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Understanding volatility as a time-varying feature of financial returns is the foundation for modern risk modelling.

□ Watch



[R Script \(https://learn.usc.edu.au/courses/35359/files/3225579?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3225579?wrap=1) 
(https://learn.usc.edu.au/courses/35359/files/3225579/download?download_frd=1)

Measuring Volatility in R

Comparing Volatility: Apple vs Microsoft

Two assets can have similar average returns but very different risk profiles. Let's compare Apple and Microsoft stock price volatility.

Step 1: Compute Monthly Returns for Apple and Microsoft

```
microsoft <- tq_get("MSFT", from = "2000-01-01") # obtain the data on MSFT
```

```
apple <- tq_get("AAPL", from = "2000-01-01") # obtain the data on AAPL
```

```
aapl_returns <- apple %>%  
  tq_transmute(  
    select = adjusted,  
    mutate_fun = periodReturn,  
    period = "monthly",
```

```

    type = "log"
  )

msft_returns <- microsoft %>%
  tq_transmute(
    select = adjusted,
    mutate_fun = periodReturn,
    period = "monthly",
    type = "log"
  )

```

Explanation:

- apple and microsoft are the price datasets downloaded earlier
- tq_transmute() converts prices into returns
- adjusted prices are used to account for dividends and splits
- period = "monthly" computes monthly returns
- type = "log" calculates continuously compounded returns

The result is two tidy time series:

- aapl_returns for Apple
- msft_returns for Microsoft

Before you proceed:

Check the contents of the two returns series

Step 2: Calculate Average Return and Volatility

```

aapl_summary <- aapl_returns %>%
  summarise(
    mean_return = mean(monthly.returns),
    volatility = sd(monthly.returns)
  )

msft_summary <- msft_returns %>%
  summarise(
    mean_return = mean(monthly.returns),
    volatility = sd(monthly.returns)
  )

```

Explanation:

- mean() computes the average monthly return
- sd() computes the standard deviation of returns
- Volatility captures the typical size of return fluctuations

We now have:

- Apple's average return and volatility
- Microsoft's average return and volatility

Step 3: Combine Apple and Microsoft Volatility Measures

To compare the two assets, we place their summary statistics into a single table.

```
volatility_comparison <- data.frame(  
  asset = c("Apple (AAPL)", "Microsoft (MSFT)"),  
  mean_return = c(  
    mean(aapl_returns$monthly.returns),  
    mean(msft_returns$monthly.returns)  
  ),  
  volatility = c(  
    sd(aapl_returns$monthly.returns),  
    sd(msft_returns$monthly.returns)  
  )  
)  
volatility_comparison
```

asset	mean_return	volatility
Apple (AAPL)	0.0183	0.114
Microsoft (MSFT)	0.0078	0.0798

Explanation:

Each row represents one asset

- mean_return summarises average performance
- volatility summarises risk
- this table allows direct comparison on a common scale

Step 4: Visualise Volatility Differences

A simple bar chart makes the risk comparison clear.

```
ggplot(volatility_comparison, aes(x = asset, y = volatility)) +  
  geom_col() +
```

```
labs(  
  
  title = "Comparison of Monthly Volatility",  
  
  x = "",  
  
  y = "Standard Deviation of Monthly Returns"  
  
) +  
  
theme_bw()
```

Explanation:

- `geom_col()` creates a bar chart
- Taller bars indicate higher volatility
- This visual highlights differences in uncertainty, not returns

Interpretation

- If one asset has higher volatility, its returns fluctuate more widely
- Higher volatility implies greater uncertainty and higher risk
- Lower volatility implies more stable, predictable outcomes

This comparison reinforces a central principle of financial analytics:

Risk is not about whether returns are positive or negative.

Risk is about how uncertain those returns are.

Mean Reversion in Volatility

Volatility tends to cluster in the short run, but over longer horizons it moves back toward a typical level. Extremely high or unusually low volatility does not persist indefinitely. Instead, it gradually drifts toward its historical average. This tendency toward normalization helps anchor financial markets and supports long-term investment decisions even when short-term conditions are unstable.

When a shock causes volatility to surge, the elevated uncertainty usually fades over time rather than remaining permanently high. Likewise, extended periods of calm are rarely permanent and are eventually followed by renewed fluctuations. These patterns suggest that volatility responds to shocks but does not spiral endlessly. Instead, it adjusts and settles, reflecting the adaptive and stabilizing forces within financial markets.

Risk–Return Trade-off

The results illustrate a classic risk–return trade-off:

- Apple offers **higher average returns**, but those returns are accompanied by **greater uncertainty**
- Microsoft offers **lower average returns**, but with **more stable and predictable outcomes**

An investor's preference between the two depends on their risk tolerance. A risk-tolerant investor may prefer Apple's higher expected return despite higher volatility, while a risk-averse investor may prefer Microsoft's lower but more stable return profile.